mitti-netra_ORDERING

MITTI-NETRA soil + structure node: ordering guide (what to buy, where, in what order)

> Companion to MANUFACTURING_MANUAL.md. Main caveat for this project: modem is a plug-in breakout. Prices and stock are from LCSC on the run date; vendors, shipping and customs rules change: verify before paying. General rules and vendor list: manuals/ORDERING_GUIDE.md.

1. Order of work

1. Read the manual and the caveat above.
2. Order the circuit boards and SMD assembly (section 2).
3. While the boards travel, order the other items (sections 3 and 5) and make the printed or laser-cut parts.
4. Assemble, program and test only after everything has arrived; follow the manual.

2. Order the circuit boards (JLCPCB, PCB + assembly)

Files (all in this repository):

- Gerbers: `utility10/pcb/release/mitti/1_fabrication/mitti_netra_gerbers.zip`
- Assembly BOM: `utility10/pcb/release/mitti/2_assembly/mitti_netra_JLC_BOM.csv`
- Pick-and-place (CPL): `utility10/pcb/release/mitti/2_assembly/mitti_netra_JLC_CPL.csv`
- Full BOM with LCSC numbers and prices: `costing/mitti_priced_BOM.csv`

Steps:

1. Create an account at **jlcpcb.com** and click *Order now*. Upload the Gerber ZIP. The viewer must show the board outline, both copper layers, silkscreen and drill holes.
2. Settings: **2 layers, 1.6 mm, FR-4**, lead-free HASL (use **ENIG** if the board has an LGA/QFN fine-pitch part, e.g. ADXL345, ESP32 module pad, BQ25570), green mask, white silk, quantity 5 (the minimum), *Remove order number: specify a location* if you want to avoid the order-number mark. Heavier copper (2 oz) only where the board notes say so.
3. Switch on **PCB Assembly**: *Economic*, assembly side *Top*, and set tooling holes *Added by JLCPCB* unless the board already has them. Choose the number of boards to assemble (2 or 5).
4. Upload the BOM and CPL. In the parts step, every line must match (LCSC numbers are in the BOM). Where a part is shown as *out of stock* or *no match*, use the alternative suggested by JLCPCB only after checking footprint and value against the list below; do not accept a different footprint.
5. In the **component placement preview**, check pin 1 / polarity of every IC, diode, LED, electrolytic capacitor, USB-C and connector against `*_assembly_top.pdf` in the same folder. Wrong rotation is the most common assembly error: ask JLCPCB support to rotate before approving.
6. Pay, then approve the assembly engineering review messages (they usually ask about a part or rotation). Delivery by DHL/FedEx/UPS takes about 1 to 2 weeks to India.

Parts **not** assembled by JLCPCB (you buy and solder them yourself):

Refs	Value	Part	LCSC
J2	Solar panel 5-6 V	JST B2B-PH-K-S(LF)(SN)	C131337
J3	Li-ion cell (protected!)	JST B2B-PH-K-S(LF)(SN)	C131337
J6	Cellular modem (SIM7080G breakout)	HCTL PZ254-2-03-Z-8.5	C2894964
J4	Strain-gauge bridge E+ E- A+ A-	JST B4B-PH-K-S(LF)(SN)	C131334
J5	Soil probe (capacitive, 3.3 V)	JST B3B-PH-K-S(LF)(SN)	C131339
J7	DS18B20 temp	JST B3B-PH-K-S	C131339
J8	UART flash/debug (USB-serial adapter)	JST B3B-PH-K-S(LF)(SN)	C131339
TP1, TP2, TP3	TP		

Order these with the same LCSC order (add them to the cart so one shipment arrives) or buy locally in India (section 5).

Parts cost and stock (live LCSC prices on the run date)

Total parts per board: **US12.97*for*1*board* * *, * * *US*11.48 each for 10 boards**, **US$9.29 each for 100 boards** (about INR 1141 / 1010 / 818 at 88 INR/USD). The per-line break-up is in `costing/mitti_priced_BOM.csv`. This is parts only: PCB fabrication, assembly fees, shipping, customs and GST come on top (see `manuals/ORDERING_GUIDE.md`, section 6).

Stock warnings on the run date:

- TP (no lcsc price (hand-fit or not priced), stock)

Refs	Qty	Value	Part	LCSC	Unit USD (1 board)	Line USD
U1	1	ESP32-S3-WROOM-1-N8	ESP32-S3-WROOM-1-N8	C2913198	4.9100	4.910
C37,C5,C8,C9,C10,C11,C12,C14,C15,C16	10	100n	Samsung CL10B104KB8NNNC 0603 100nF X7R 50V	C1591	0.0075	0.075
C38,C1,C2,C4,C13	5	10u	0805 10uF X5R	C15850	0.0441	0.221
R42	1	10k	0603 10k 1%	C25804	0.0024	0.002
C39,C3	2	1u	0603 1uF X7R	C15849	0.0144	0.029
SW2	1	Reset (EN)	Omron B3U-1000P	C231329	0.1894	0.189
SW3	1	Boot (IO0)	Omron B3U-1000P	C231329	0.1894	0.189
J1	1	USB-C (charge + program)	TYPE-C-31-M-12	C165948	0.1890	0.189
R43,R44	2	5.1k	0603 5.1k 1%	C23186	0.0016	0.003
J2	1	Solar panel 5-6 V	JST B2B-PH-K-S(LF)(SN)	C131337	0.0359	0.036
D1,D2	2	SS14	SS14	C84343	0.0252	0.050
U4	1	TP4056	TPOWER TP4056	C382139	0.1208	0.121
R1	1	2.2k	0603 2.2k 1%	C4190	0.0018	0.002
R16	1	6.2k	UNI-ROYAL 0603WAF6201T5E	C4260	0.0023	0.002

Refs	Qty	Value	Part	LCSC	Unit USD (1 board)	Line USD
TH1	1	NTC 10k B3950	FH CMFB103F3950FANT	C51597	0.0785	0.079
D3	1	Red	Hubei KENTO KT-0603R red (Everlight 19-217 was out of stock)	C2286	0.0077	0.008
R2,R3,R15,R14	4	1k	0603 1k 1%	C21190	0.0017	0.007
D4,D5	2	Green	Hubei KENTO KT-0603YG yellow-green 570 nm, Vf about 2 V (lights well from 3.3 V with 1 k; the Everlight 19-217 green was out of stock)	C2289	0.0105	0.021
J3	1	Li-ion cell (protected!)	JST B2B-PH-K-S(LF)(SN)	C131337	0.0359	0.036
U3	1	AP2112K-3.3	AP2112K-3.3TRG1	C51118	0.1814	0.181
R4,R5	2	330k	UNI-ROYAL 0603WAF3303T5E	C23137	0.0016	0.003
Q1	1	AO3401A	AO3401A	C15127	0.0969	0.097
R6,R8	2	100k	0603 100k 1%	C25803	0.0030	0.006
Q2	1	AO3400A	AO3400A	C20917	0.0899	0.090
R7,R11,R12	3	100	UNI-ROYAL 0603WAF1000T5E	C22775	0.0024	0.007
C6	1	100u/35V	Lelon VZH101M1VTR-0607	C176683	0.0881	0.088
C7	1	22u/10V	Samsung CL21A226MPQNNNE	C29277	0.0377	0.038
J6	1	Cellular modem (SIM7080G breakout)	HCTL PZ254-2-03-Z-8.5	C2894964	0.0766	0.077
U5	1	ADXL345	ADXL345BCCZ-RL7	C9667	5.4131	5.413
R9,R10,R13	3	4.7k	0603 4.7k 1%	C23162	0.0028	0.008
U2	1	HX711	HX711	C43656	0.6211	0.621
J4	1	Strain-gauge bridge E+ E- A+ A-	JST B4B-PH-K-S(LF)(SN)	C131334	0.0458	0.046
J5	1	Soil probe (capacitive, 3.3 V)	JST B3B-PH-K-S(LF)(SN)	C131339	0.0420	0.042
J7	1	DS18B20 temp	JST B3B-PH-K-S	C131339	0.0420	0.042
J8	1	UART flash/debug (USB-serial adapter)	JST B3B-PH-K-S(LF)(SN)	C131339	0.0420	0.042
TP1,TP2,TP3	3	TP				

2b. Enclosure: printed parts and fasteners (generated from the board)

- Print `4_enclosure/*_enclosure_plate.stl` (106 x 96 x 3 mm, PETG or ABS). Buy an **IP65 ABS junction box** with at least 110 x 100 x 38 mm inside, cable glands and a breather/vent membrane; fix the plate with 4 x M4 screws.
- Board to plate: 4 x M2.5 x 6 mm self-tapping screws. field node: mount on this plate inside an IP65 ABS junction box with cable glands and a vent membrane; place the 18650 holder and the modem breakout in the same box; keep the NTC TH1 side of the board next to the cell; shade the box from direct sun

3. Bought-in and mechanical items

Bought-in

SIM7080G (LTE-M/NB-IoT) breakout board with nano-SIM holder and antenna; protected 18650 cell (or protected pouch cell) with JST-PH 2 lead; 5-6 V solar panel (2 W) with JST-PH 2 lead; capacitive soil moisture probe (3.3 V, analog) with JST-PH 3 lead; DS18B20 waterproof probe; strain gauges (350 ohm, GF about 2) with bridge completion; IP65 enclosure with cable glands; a SIM and data plan.

4. Confidence (read before you order)

- **Order confidence: High**: all parts live-checked, none flagged.
- **Works-as-intended confidence: Medium**: Charger thermal pad now grounded with thermal vias; NTC window stops Li-ion charging outside about 6 to 44 C; power diodes corrected; IP65 box with a printed mounting plate; firmware core 23 checks and the pin map match; the cellular modem is a plug-in breakout and the ESP32 firmware compiles for the ESP32-S3 with 0 warnings (not yet run on hardware).
- **First thing to do:** Bring up power and charging first (check TEMP voltage at room temperature), then the modem last.
- Rubric and the reasons for every project: `costing/CONFIDENCE.md`. These are judgements, not measurements: nothing here has been built.

5. Where to buy and whom to approach

Item type	India (fast, GST invoice)	Overseas (cheaper, 1 to 3 weeks, customs)
Printed circuit board and SMD assembly	JLCPCB is the practical choice for 5 to 100 boards; local PCB shops quote per project (ask for the Gerber ZIP price)	**JLCPCB** (jlcpcb.com), PCBWay (pcbway.com)
Connectors, headers, passives, ICs by LCSC number	element14 India, Mouser India, DigiKey India (genuine, GST invoice, higher price); local electronics markets for common parts	**LCSC** (lcsc.com) with the same cart as JLCPCB
Modules: ESP32 dev boards, SIM7080G, LoRa (RA-08 / RAK3172), pH / EC / DO modules, relay modules, load cells, ultrasonic sensors	Robu.in, Robocraze, ThinkRobotics, Sunrom Electronics, Amazon.in	AliExpress, Seeed Studio, Waveshare
Motors, stepper drivers, wheels	Robu.in, Robocraze, local robotics shops	StepperOnline (omc-stepperonline.com)
Li-ion / LiFePO4 cells and packs, solar panels, MPPT controllers	Authorised battery and solar dealers, Amazon.in (buy protected cells with an invoice)	not recommended by post (shipping rules for batteries)
Laser-cut and 3D-printed parts	Local laser-cutting and 3D-printing shops (search IndiaMART, Justdial or maker spaces): send the SVG/DXF/STL from the repository and ask for a quote per piece	JLC3DP (jlc3dp.com), PCBWay 3D printing

Item type	India (fast, GST invoice)	Overseas (cheaper, 1 to 3 weeks, customs)
Fasteners, tubing, hardware, enclosures	Local hardware, Robu.in, Amazon.in	n/a

How to approach a vendor: send the project name, the exact file (BOM, CPL, Gerber ZIP, STL or SVG), the quantity, and ask for the price, the lead time, the GST invoice and the warranty or return terms. Keep the reply for your records.